EXERCISES 4.1

Answers to selected odd-numbered problems begin on page ANS-12.

4.1.1 Complex Exponential Function

In Problems 1–4, find the derivative f' of the given function f.

1. $f(z) = z^2 e^{z+i}$

2. $f(z) = \dfrac{3e^{2z} - ie^{-z}}{z^3 - 1 + i}$

3. $f(z) = e^{iz} - e^{-iz}$

4. $f(z) = ie^{1/z}$

In Problems 5–8, write the given expression in terms of x and y.

5. $\left|e^{z^2 - z}\right|$

6. $\arg(e^{z - i/z})$

7. $\arg(e^{i(z+\bar{z})})$

8. $\overline{ie^z + 1}$

In Problems 9–12, express the given function f in the form $f(z) = u(x, y) + iv(x, y)$.

9. $f(z) = e^{-iz}$

10. $f(z) = e^{2\bar{z}+i}$

11. $f(z) = e^{z^2}$

12. $f(z) = e^{1/z}$

13. Use the sufficient conditions for differentiability from Section 3.3 to determine where the function $f(z) = e^{2\bar{z}+i}$ is differentiable.

14. Use the sufficient conditions for differentiability from Section 3.3 to determine where the function $f(z) = e^{z^2}$ is differentiable.

In Problems 15–20, find the image of the given set under the exponential mapping.

15. The line $y = -2$.

16. The line $x = 3$.

17. The infinite strip $1 < x \leq 2$.

18. The square with vertices at 0, 1, $1 + i$, and i.

19. The rectangle $0 \leq x \leq \log_e 2$, $-\pi/4 \leq y \leq \pi/2$.

20. The semi-infinite strip $-\infty < x \leq 0$, $0 \leq y \leq \pi$.

4.1.2 Complex Logarithmic Function

In Problems 21–26, find all complex values of the given logarithm.

21. $\ln(-5)$

22. $\ln(-ei)$

23. $\ln(-2 + 2i)$

24. $\ln(1 + i)$

25. $\ln\left(\sqrt{2} + \sqrt{6}i\right)$

26. $\ln\left(-\sqrt{3} + i\right)$

In Problems 27–32, write the principal value of the logarithm in the form $a + ib$.

27. $\text{Ln}(6 - 6i)$

28. $\text{Ln}\left(-e^2\right)$

29. $\text{Ln}(-12 + 5i)$

30. $\text{Ln}(3 - 4i)$

31. $\text{Ln}\left[\left(1 + \sqrt{3}i\right)^5\right]$

32. $\text{Ln}\left[(1 + i)^4\right]$

In Problems 33–36, find all complex values of z satisfying the given equation.

33. $e^z = 4i$

34. $e^{1/z} = -1$

35. $e^{z-1} = -ie^3$

36. $e^{2z} + e^z + 1 = 0$

In Problems 37–40, find a domain in which the given function f is differentiable; then find the derivative f'.

37. $f(z) = 3z^2 - e^{2iz} + i\operatorname{Ln} z$

38. $f(z) = (z+1)\operatorname{Ln} z$

39. $f(z) = \dfrac{\operatorname{Ln}(2z-i)}{z^2+1}$

40. $f(z) = \operatorname{Ln}\left(z^2+1\right)$

In Problems 41–46, find the image of the given set under the mapping $w = \operatorname{Ln} z$.

41. The ray $\arg(z) = \pi/6$.

42. The positive y-axis.

43. The circle $|z| = 4$.

44. The region in the first quadrant bounded by the circles $|z| = 1$ and $|z| = e$.

45. The annulus $3 \le |z| \le 5$.

46. The region outside the unit circle $|z| = 1$ and between the rays $\arg(z) = \pi/4$ and $\arg(z) = 3\pi/4$.

Focus on Concepts

47. Use (1) to prove that $e^{z_1}/e^{z_2} = e^{z_1 - z_2}$.

48. Use (1) and de Moivre's formula to prove that $(e^{z_1})^n = e^{nz_1}$, n an integer.

49. Determine where the complex function $e^{\bar{z}}$ is analytic.

50. In this problem, we will show that the complex exponential function defined by (1) is the only complex entire function f that agrees with the real exponential function e^x when z is real and that has the property $f'(z) = f(z)$ for all z.

(**a**) Assume that $f(z) = u(x,\ y) + iv(x,\ y)$ is an entire complex function for which $f'(z) = f(z)$. Explain why u and v satisfy the differential equations:

$$u_x(x,\ y) = u(x,\ y) \quad \text{and} \quad v_x(x,\ y) = v(x,\ y).$$

(**b**) Show that $u(x,\ y) = a(y)e^x$ and $v(x,\ y) = b(y)e^x$ are solutions to the differential equations in (a).

(**c**) Explain why the assumption that $f(z)$ agrees with the real exponential function for z real implies that $a(0) = 1$ and $b(0) = 0$.

(**d**) Explain why the functions $a(y)$ and $b(y)$ satisfy the system of differential equations:

$$a(y) - b'(y) = 0$$
$$a'(y) + b(y) = 0.$$

(**e**) Solve the system of differential equations in (d) subject to the initial conditions $a(0) = 1$ and $b(0) = 0$.

(**f**) Use parts (a)–(e) to show that the complex exponential function defined by (1) is the only complex entire function $f(z)$ that agrees with the real exponential function when z is real and that has the property $f'(z) = f(z)$ for all z.

51. Describe the image of the line $y = x$ under the exponential function. [*Hint*: Find a polar expression $r(\theta)$ of the image.]

52. Prove that e^z is a one-to-one function on the fundamental region $-\infty < x < \infty$, $-\pi < y \le \pi$.

53. Prove that $\ln\left(\dfrac{z_1}{z_2}\right) = \ln z_1 - \ln z_2$ for all nonzero complex numbers z_1 and z_2.

54. Prove that $\ln z_1^n = n \ln z_1$ for all nonzero complex numbers z_1 and all integers n.

55. (**a**) Find two nonzero complex numbers z_1 and z_2 so that $\operatorname{Ln}(z_1 z_2) \neq \operatorname{Ln} z_1 + \operatorname{Ln} z_2$.

(**b**) Find two nonzero complex numbers z_1 and z_2 so that $\operatorname{Ln}(z_1 z_2) = \operatorname{Ln} z_1 + \operatorname{Ln} z_2$.

(**c**) What must be true about z_1 and z_2 if $\operatorname{Ln}(z_1 z_2) = \operatorname{Ln} z_1 + \operatorname{Ln} z_2$?

56. Is $\operatorname{Ln} z_1^n = n \operatorname{Ln} z_1$ for all integers n and all nonzero complex numbers z_1? Defend your position with a short proof or a counterexample.

Computer Lab Assignments

Most CASs have a built in function to evaluate $\operatorname{Ln} z$. For example, in *Mathematica*, the syntax **Log[a + b I]** finds the principal value of the complex logarithm of $a + bi$. To find a numerical approximation of this value enter the expression **N[Log[a + b I]]**. For example, *Mathematica* indicates that **N[Log[2 + 3 I]]** is approximately $1.28247 + 0.982794i$.

In Problems 57–62, use a CAS to compute $\operatorname{Ln} z$.

57. $z = -1 - i$

58. $z = 2 - 3i$

59. $z = 3 + \pi i$

60. $z = 13 + \sqrt{2}i$

61. $z = 4 + 10i$

62. $z = \dfrac{12 - i}{2 + 3i}$

In Problems 63–66, use a CAS to find one solution to the equation.

63. $e^{5z-i} = 12i$

64. $e^{iz} = 2 - 5i$

65. $3e^{(2+i)z} = 5 - i$

66. $ie^{z-2} = \pi$

4.2 Complex Powers

To The Instructor : In this section, we study complex powers z^α where α is a complex constant. This section can be omitted without significantly affecting the development of topics in later chapters.

INTRODUCTION In Section 2.4 we examined special power functions of the form z^n and $z^{1/n}$ for n an integer and $n \geq 2$. These functions generalized the usual squaring, cubing, square root, cube root, and so on, functions of elementary calculus. In this section we analyze the problem of raising a complex number to an arbitrary real or complex power. For simple integer powers this process is easily understood in terms of complex multiplication. For example, $(1+i)^3 = (1+i)(1+i)(1+i) = -2+2i$. However, there is no similar description for a complex power such as $(1+i)^i$. In order to define such expressions, we will use the complex exponential and logarithmic functions from Section 4.1.

Complex Powers

Complex powers, such as $(1+i)^i$ mentioned in the introduction, are defined in terms of the complex exponential and logarithmic functions. Recall from (10) in Section 4.1 that $z = e^{\ln z}$ for all nonzero complex numbers z. Thus, when n is an integer it follows from Theorem 4.1.2(*iv*) that z^n can be written as $z^n = \left(e^{\ln z}\right)^n = e^{n \ln z}$. This formula, which holds for integer exponents n, suggests the following formula used to define the complex power z^α for any complex exponent α.

DEFINITION 4.2.1 Complex Powers

If α is a complex number and $z \neq 0$, then the **complex power** z^α is defined to be:

$$z^\alpha = e^{\alpha \ln z}. \tag{1}$$